Triptans (serotonin, 5-HT1B/1D agonists) in migraine: detailed results and methods of a meta-analysis of 53 trials.

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The triptans, selective serotonin 5-HT1B/1D agonists, are very effective acute migraine drugs. Soon, seven different triptans will be clinically available at 13 different oral doses, making evidence-based selection guidelines necessary. Triptan trials have similar designs, facilitating meta-analysis. We wished to provide an evidence-based foundation for using triptans in clinical practice, and to review the methodological issues surrounding triptan trials. We asked pharmaceutical companies and the principal investigators of company-independent trials for the 'raw patient data' of all double-blind, randomized, controlled, clinical trials with oral triptans in migraine. All data were cross-checked with published or presented data. We calculated summary estimates across studies for important efficacy and tolerability parameters, and compared these with those from direct, head-to-head, comparator trials. Out of 76 eligible clinical trials, 53 (12 not yet published) involving 24089 patients met the criteria for inclusion. Mean results (and 95% confidence intervals) for sumatriptan 100 mg, the first available and most widely prescribed oral triptan, are 59% (57-60) for 2 h headache response (improvement from moderate or severe to mild or no pain); 29% (27-30) for 2 h pain free (improvement to no pain); 20% (18-21) for sustained pain free (pain free by 2 h and no headache recurrence or use of rescue medication 2-24 h post-dose), and 67% (63-70) for consistency (response in at least two out of three treated attacks); placebo-subtracted proportions for patients with at least one adverse event (AE) are 13% (8-18), for at least one central nervous system AE 6% (3-9), and for at least one chest AE 1.9% (1.0-2.7). Compared with these data: rizatriptan 10 mg shows better efficacy and consistency, and similar tolerability; eletriptan 80 mg shows better efficacy, similar consistency, but lower tolerability; almotriptan 12.5 mg shows similar efficacy at 2 h but better sustained pain-free response, consistency, and tolerability; sumatriptan 25 mg, naratriptan 2.5 mg and eletriptan 20 mg show lower efficacy and better tolerability; zolmitriptan 2.5 mg and 5 mg, eletriptan 40 mg, and rizatriptan 5 mg show very similar results. The results of the 22 trials that directly compared triptans show the same overall pattern. We received no data on frovatriptan, but publicly available data suggest substantially lower efficacy. The major methodological issues involve the choice of the primary endpoint, consistency over multiple attacks, how to evaluate headache recurrence, use of placebo-subtracted proportions to control for across-study differences, and the difference between tolerability and safety. In addition, there are a number of methodological issues specific for direct comparator trials, including encapsulation and patient selection. At marketed doses, all oral triptans are effective and well tolerated. Differences among them are in general relatively small, but clinically relevant for individual patients. Rizatriptan 10 mg, eletriptan 80

mg and almotriptan 12.5 mg provide the highest likelihood of consistent success. Sumatriptan features the longest clinical experience and the widest range of formulations. All triptans are contra-indicated in the presence of cardiovascular disease.